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Pilot Certified private pilot of light aircrafts.

Education

Year	Degree	University	CGPA
2017- present	Ph.D.(Aerospace Engineering)	Texas A&M university-USA	4/4
2014 - 2016	M.Tech.(Aerospace Engineering)	Indian Institute of Technology-Kanpur, India	10/10
2010- 2014	B.E.(Aeronautical Engineering)	Madras Institute of Technology, Anna University, Chennai, India	9.4/10

Research/Project Experience

Aug 2017 - Present Texas A&M university, USA
PhD (work in progress): Computational ductile fracture
Advisor: Dr. Amine Benzerga

- This research aims to formulate a constitutive theory and develop a robust code based thereof, enabling ductile fracture simulations under complex loadings, such as torsion and compression. Over the past decade, the ductile fracture community has made significant progress in developing constitutive relations for porous material plasticity from first principles. This new class of models holds the promise to completely change our understanding of some of the most challenging problems. Whereby, a stable numerical scheme of multi-surface plasticity serves to predict ductile fracture under arbitrary loadings.
- Sponsors: National Science foundation research fellowship and T3 grant.

Aug 2020 - Dec 2020 Texas A&M university, USA
Review of a multi-scale non-linear homogenization scheme

- It involves reviewing non-linear homogenization scheme in the context of multi-scale microstructure. An advanced non-linear homogenization scheme that combine: Fully-optimized second order homogenization and iterated second order homogenization, by Pontecastaneda et. al was reviewed as a part of this project.

Jan 2020 - Apr 2020 Texas A&M university, USA
On limit analysis using functional interpolation

- A class of problems in micro-mechanics involve the infimum and supremum (Inf-Sup) condition. This is the case when limit analysis is combined with (nonlinear) homogenization of porous media. Therefore, we minimize/maximize some functional over a set of fields that obey certain constraints, namely kinematic admissibility, static admissibility, and/or incompressibility.

Aug 2018 - Texas A&M university, USA
Dec 2018 *Void growth and coalescence in porous single crystals*

- Unit cell calculations on the porous crystal material was carried out in ABAQUS using a crystal plasticity umat. The effect of triaxiality on void growth and strain to coalescence were studied. The failure locus under non-proportional loading paths was analyzed.

July 2016 - VEM Technologies Pvt. Ltd., Hyderabad, India
July 2017 *Engineer*

- The brief description of the activities include: Computation of loads on Aircraft structures, such as wings/fuselage, to optimize the strength based on the static and dynamic analysis using the FE package ANSYS.

May 2015 - Indian Institute of Technology-Kanpur, India
May 2016 *Master's Thesis: Numerical assessment of GLD plasticity model for porous materials*
 Advisors: Dr. P.M.Mohite, Dr. C.S.Upadhyay

- This work focused on studying the behavior of Elasto-plastic materials under finite deformation. The plasticity behavior of a material was simulated using the GLD (Gologanu-Leblond-Devaux) model. The model was implemented in ABAQUS using a user subroutine (UMAT) for a complete 3D plasticity analysis. The problems such as ductile fracture of round tensile bar and plate with a hole were studied.

Jan 2015 - Indian Institute of Technology-Kanpur, India
Apr 2015 *Finite element code development*

- FE code was developed from scratch in FORTRAN for solving generalized second order elliptic equation using hp-method which could work out a range of problems, including, structural analysis of bar, heat conduction in plate, diffusion-convection near boundary layer.

Jan 2015 - Indian Institute of Technology-Kanpur, India
Apr 2015 *Micromechanics project in composites*

- The project involved estimating the Engineering constants of a Carbon-Epoxy lamina, using various micromechanical approaches. The bounds and estimators of the effective anisotropic material by different models were analyzed.

Dec 2013 - Madras Institute of Technology and IIT-Madras
Apr 2014 *UG Dissertation: Optimisation of Ultrasonic transducer staggered configuration for detection of cracks*
 Advisors: Dr. R. Dhanaraj, Dr. H.S.N. Murthy

- The time-of-flight diffraction technique was used to optimize the ultrasonic transmitter-receiver configurations for monitoring crack propagation during cyclic loading. The simulation was carried out by using finite element model using ANSYS software and the results are compared with Finite-difference time-domain (FDTD) scheme.

May 2013 - Indian Institute of Technology-Madras
Dec 2013 *Contact Analysis of 2D structures*
 Advisor: Dr. H.S.N. Murthy

- Finite element code was developed to simulate frictionless contact between elastic bodies, sphere and half space. This work was conducted during summer fellowship programme at IITM.

Dec 2012 - Madras Institute of Technology, Chennai, India

Nov 2013 *25 Seater Aircraft design project.*

- As a team leader of the project, the design of Aerodynamics and Structures of a 25 seater aircraft were carried out. It was a preliminary design on paper based on the similar aircrafts of the same category designed in the past .

Teaching Experience

Aug 2018 - Teaching Assistant in Dept. of Aerospace Engineering, Texas A&M university, USA

Dec 2018 *AERO306-Aerospace structural analysis*

- As a teaching assistant, the queries in the concepts, problem solving methodology were explained, quizzes, assignments, exams were graded.

Jul 2015 - Teaching Assistant in Dept. of Aerospace Engineering, Indian Institute of Technology-Kanpur, India

Dec 2015 *AE451-Experiments in Aerospace Engineering*

- Lectured the concepts behind the experiment on 'Combined loading of Aluminium alloy' and graded the student's reports.

Honors and Awards

■ Aero graduate Excellence Fellowship award

for the Academic Excellence in the grad school at Texas A&M university.

■ The Certificate of Merit

for the Academic Excellence in The Master of Technology in Aerospace Engineering at IIT-Kanpur.

■ Gold medalist

in B.E. Aeronautical Engineering at Madras Institute of Technology, Anna university.

■ Best outgoing student

Dr.A.Narasinga Rao prize for the best outgoing student of B.E/B.Tech 2013-14 at Madras Institute of Technology, Anna university.

■ State 1st in Tamilnadu, India

by securing full-marks in Physics, Chemistry, Maths, Biology in Higher Secondary School Certificate.

Co-curricular activities

- Undergraduate research mentor at TAMU: Involves coordinating and assessing research with an undergraduate student.
- Member of the club: Texas Running against cancer (TRAC) at TAMU.
- Class Representative in UG from Nov 2012-Dec 2013.
- Interacted with students of Velammal School, Surapet, Chennai, India, and helped them to overcome their struggles for their Higher Secondary School Certificate course.

References

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